

FitFuel Purchase Funnel & Marketing Analytics

Analysis period: Aug 28 – Sep 4, 2026 | Data source: GA4 export to BigQuery

Executive summary

This analysis examines on-site behavior for FitFuel's e-commerce store using raw event-level data exported from GA4 into BigQuery. The purchase funnel shows a steep decline between adding a product to cart and beginning checkout — only 4% of users who added an item to their cart went on to start checkout, the single largest leak in the entire funnel. Traffic is almost entirely first-time visitors (99.9% new), and 70% of sessions land on the homepage rather than a category or product page, suggesting limited use of targeted landing pages for acquisition. Product-level analysis identifies Resistance Bands and Whey Protein (Vanilla) as the strongest converters from view to cart.

Methodology

- Traffic was generated via a Python script simulating FitFuel site visitors, sent to Google Analytics 4, and exported automatically to BigQuery.
- All analysis was performed directly in BigQuery SQL against the raw events_* tables (event-level, not pre-aggregated).
- Unique users were counted using DISTINCT user_pseudo_id per event type.
- Visualizations were built in Google Looker Studio (Data Studio), connected to BigQuery via custom SQL.
- Limitation: this is synthetic data intended for skill-building, not real customer traffic — findings illustrate method, not real FitFuel performance.

1. Purchase funnel

Unique users reaching each stage of the buying journey, Aug 28 – Sep 4:

Funnel stage	Unique users	% of prior stage	Notes
Page view	1,776	—	Entry point
View item	1,287	72%	
Add to cart	338	26%	Sharpest early drop
Begin checkout	15	4%	Largest single leak in the funnel
Purchase (via checkout)	12	80%	Excludes 43 purchases with no checkout event (data quality issue, see below)

Key finding: the drop from add-to-cart (338) to begin-checkout (15) is the largest leak in the funnel — a 96% loss at this single step. This is the highest-leverage point for improvement: reducing checkout friction (extra steps, required account creation, unexpected costs) would have more impact than any other funnel stage.

2. Data quality finding: purchases without checkout

Raw purchase_users (55) was found to be higher than begin_checkout_users (15) — logically inconsistent, since a purchase should follow a checkout start. Investigation via a LEFT JOIN comparison found 43 of the 55 purchasers (78%) had no matching begin_checkout event at all.

Conclusion: this is a data generation artifact in the synthetic dataset (the traffic-simulation script created some purchase events independently of the checkout sequence), not a real behavioral pattern. The funnel table above uses the corrected

figure of 12 true checkout → purchase conversions. Flagging and correcting for this before drawing conclusions is a standard data-quality step in any real analysis.

3. Product-level conversion (view → add to cart)

Top 5 products by view-to-cart conversion rate:

Product page	Views	Added to cart	View → cart %
FitFuel Resistance Bands	72	53	73.6%
FitFuel Whey Protein — Vanilla	58	35	60.3%
FitFuel Wellness Bundle	74	43	58.1%
FitFuel Whey Protein — Chocolate	56	32	57.1%
FitFuel Creatine Monohydrate	57	29	50.9%

Category landing pages (e.g. Fitness Accessories, Protein & Supplements) receive far more views than any single product, but convert to cart at a much lower rate — shoppers browse categories, then convert only after reaching a specific product page. Resistance Bands stands out with a 73.6% view-to-cart rate, well above the product average, suggesting strong product-market fit or particularly persuasive product copy worth studying and replicating.

4. Traffic composition: new vs. returning users

99.9% of sessions (1,775 of 1,776) came from new users; only 1 session was from a returning user. There is effectively no repeat-visit behavior in this dataset. In a real store, this would be a red flag for retention/remarketing strategy — it would suggest testing email capture, retargeting ads, or loyalty incentives to bring first-time visitors back.

5. Landing page distribution

70% of sessions (1,243 of 1,776) landed on the homepage rather than a category or product page. This indicates most traffic is not being directed by SEO or paid campaigns to relevant, high-intent pages — an opportunity to test category-specific landing pages for search and social campaigns instead of routing all traffic through the homepage.

Recommendations

- Prioritize reducing checkout friction — this is the single biggest lever in the entire funnel (96% drop-off).
- Investigate and promote whatever makes Resistance Bands and Whey Protein (Vanilla) convert well above average — pricing, imagery, or copy patterns worth replicating on other product pages.
- Test category- or product-specific landing pages in marketing campaigns instead of defaulting to the homepage.
- Given near-zero returning traffic, evaluate retention tactics (email capture, retargeting, loyalty offers) to convert one-time visitors into repeat customers.
- Audit event tracking/checkout instrumentation to prevent future purchase events from firing without a preceding checkout event.

Tools used

Python (synthetic data generation) · Google Analytics 4 · Google BigQuery (SQL) · Google Looker Studio (visualization)